Prime Factor Decomposition

Exam code: 0607

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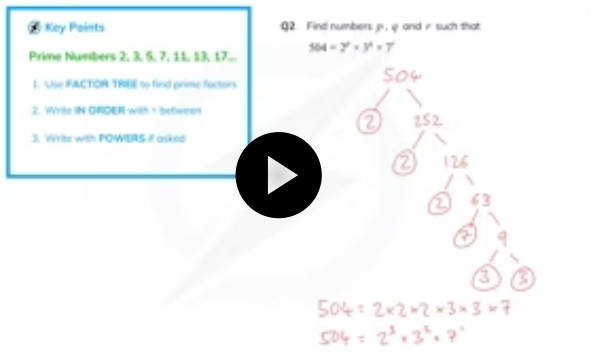

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Help me understand

Prime factor decomposition

What are prime factors?

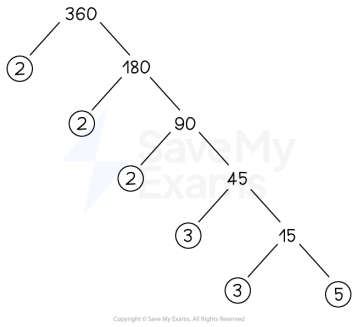
- A **factor** of a given number is a value that divides the given number exactly, with no remainder
  - e.g. 6 is a factor of 18
- A **prime** number is a number which has exactly two **factors;** itself and 1
  - e.g. 5 is a prime number, as its only factors are 5 and 1
  - You should remember the first few prime numbers:
    - 2, 3, 5, 7, 11, 13, 17, 19, ...
- The **prime factors** of a number are therefore all the primes which multiply to give that number
  - e.g. The prime factors of 30 are 2, 3, and 5
    - $2 \times 3 \times 5 = 30$

How do I find prime factors?

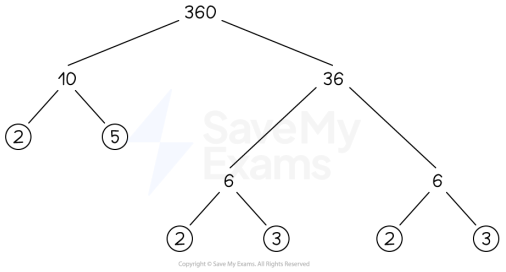
- Use a **factor tree** to find prime factors
  - Split the number up into a **pair of factors**
  - Then split each of those factors up into **another pair**
  - Continue splitting up factors along each "branch" until you get to a **prime number**
    - These can not be split into anything other than 1 and themselves

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

smallest possible prime number
  - This gives a prime factor at each stage



- Another way to split each number up by dividing by **easy numbers** such as 10
  - These do not need to be prime numbers
  - You will need to split any non-prime numbers into pairs of factors
  - These trees can get messy



- The number will always end up with the **same numbers circled** no matter which pairs of numbers you choose
- A number can be uniquely written as a **product of prime factors**
  - Write the prime factors as a multiplication, in ascending order
    - $360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5$
  - This can then be written more concisely using powers
    - $360 = 2^3 \times 3^2 \times 5$

Examiner Tips and Tricks

Exam questions on this topic will ask you to express a number as a product of its prime factors. You should simplify your final answer by using indices. You need to show your working clearly to gain full marks.

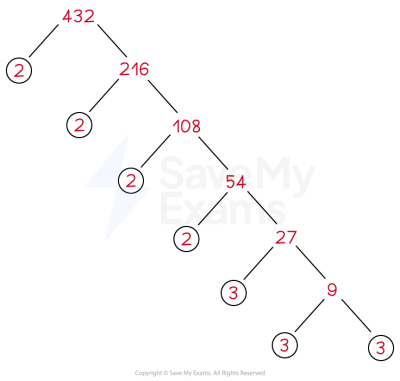
working clearly.

Answer:

Create a factor tree
Start with 432 and choose any two numbers that multiply together to make 432

- 2 and 216 is an obvious pair

Continue to split non-prime numbers into pairs of factors



The answer will be the same regardless of the factors chosen in the first step

Write the prime numbers out as a product

$$432 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$$

Any repeated prime factors can be written as a power

$$432 = 2^4 \times 3^3$$

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